

SEMICONDUCTOR TRADE ANALYSIS

U.S. Semiconductor Exports — 2025 | Part 2 Documentation

Overview

This document describes the SQL queries developed to analyze U.S. semiconductor trade data stored in the `semiconductors_db` MySQL database. The `exports_data` table (950 rows) holds partner-level monthly shipment records including FOB values, continent assignments, and partner ISO codes. The `imports_data` table (373 rows) mirrors this structure for inbound flows. Together, the eight queries documented here cover market concentration, trade volumes, geographic distribution, temporal patterns, volatility, and competitive benchmarking.

Query 1 — Market Concentration Index (HHI) for Imports

Objective

Compute the Herfindahl–Hirschman Index (HHI) for U.S. semiconductor imports in order to assess how concentrated import sourcing is across trading partners. An HHI below 0.15 indicates a competitive, diversified supply base; values above 0.25 signal high concentration.

SQL Code

```
WITH partner_shares AS (  
    SELECT reporterDesc,  
           SUM(cifvalue) AS total_imports,  
           SUM(cifvalue) * 1.0 / (SELECT SUM(cifvalue)  
                                  FROM imports_data) AS shares  
    FROM   imports_data  
    GROUP BY reporterDesc  
)  
SELECT SUM(POWER(shares, 2)) AS hhi  
FROM   partner_shares;
```

Result

Metric	Value
hhi	0.6971088291781666

Key Design Choices

- A Common Table Expression (CTE) named `partner_shares` calculates each reporter's share of total CIF value in a single pass, avoiding a correlated subquery inside the GROUP BY.
- Multiplying by 1.0 before the division forces floating-point arithmetic, preventing integer truncation in MySQL.
- `POWER(shares, 2)` squares each market share; SUM aggregates them into the final HHI scalar.
- An HHI of ~0.70 signals extreme import concentration — the U.S. relies overwhelmingly on a very small number of source countries for semiconductors.

Troubleshooting Notes

An earlier attempt grouped by `reporterDesc` inside the SELECT list without including it in GROUP BY, triggering Error 1055 (non-aggregated column). The fix was to restructure the calculation inside the CTE so the outer SELECT only reads the pre-aggregated shares column.

Query 2 — Market Concentration Index (HHI) for Exports

Objective

Apply the same HHI methodology to U.S. semiconductor exports, using FOB values and grouping by destination partner ISO code. This allows a direct comparison of supply-side versus demand-side concentration.

SQL Code

```
WITH partner_shares AS (
  SELECT partnerISO,
         SUM(fobvalue) AS total_exports,
         SUM(fobvalue) * 1.0 / (SELECT SUM(fobvalue)
                                FROM exports_data) AS shares
  FROM   exports_data
  GROUP BY partnerISO
)
SELECT SUM(POWER(shares, 2)) AS hhi
FROM   partner_shares;
```

Result

Metric	Value
hhi	0.17111146110000003

Key Design Choices

- partnerISO is used as the grouping key rather than a full country name, ensuring consistent grouping even when partner descriptions vary across rows.
- The exports HHI (~0.17) is substantially lower than the imports HHI (~0.70), indicating that U.S. semiconductor exports are distributed across a broader set of destination markets.
- The structural symmetry between this query and Query 1 makes the two results directly comparable.

Troubleshooting Notes

Two earlier attempts failed with Error 1111 (invalid use of group function) when nesting SUM inside POWER directly. Moving the share calculation into a CTE resolved the nesting issue. A third attempt failed with Error 1054 (unknown column 'share') because the alias was misspelled; renaming the CTE column to shares corrected it.

Query 3 — Total FOB Value by Partner (Exports)

Objective

Rank all destination countries by total FOB export value to identify the top ten markets receiving U.S. semiconductor shipments.

SQL Code

```
SELECT partnerISO,
       SUM(fobvalue) AS total_fob
FROM   Exports_data
GROUP  BY partnerISO
ORDER  BY total_fob DESC
LIMIT 10;
```

Result

Partner	Total FOB (USD)
---------	-----------------

Mexico	8,534,846,870
China	6,196,381,769
Malaysia	4,665,345,607
Other Asia, nes	2,424,024,977
Viet Nam	1,164,779,997
South Korea	1,130,306,889
Brazil	842,205,357
Ireland	437,111,164
Canada	395,226,100
Singapore	384,311,392

Key Design Choices

- ORDER BY total_fob DESC with LIMIT 10 efficiently surfaces only the top recipients, keeping the result set focused for visualization and reporting.
- Mexico is the dominant export destination, receiving roughly 38% more than China and nearly twice the value shipped to Malaysia.
- The presence of 'Other Asia, nes' (not elsewhere specified) in fourth place suggests a data-quality flag worth monitoring in future cleansing cycles.

Query 4 — Monthly Exports by Partner

Objective

Decompose export volumes by both partner country and calendar month, returning the full cross-product sorted by export value. This granular view supports time-series analysis and seasonal pattern detection.

SQL Code

```
SELECT partnerISO,
       refMonth,
       SUM(fobvalue) AS monthly_exports
FROM   Exports_data
GROUP  BY partnerISO, refMonth
ORDER  BY monthly_exports DESC;
```

Sample Results (top rows)

Partner	Month	Monthly Exports (USD)
Mexico	April	1,016,946,152
Mexico	January	970,336,727
Mexico	October	945,354,112
Mexico	March	920,709,486
Mexico	July	901,075,475
Mexico	February	844,064,330
China	January	838,723,007
Mexico	May	828,758,946

Key Design Choices

- Grouping on two dimensions (partnerISO, refMonth) produces one row per country-month combination, enabling downstream pivot or chart generation.
- 950 total rows are returned (all partner-month pairs), giving complete coverage without filtering.
- Mexico's April peak (~\$1.02 B) stands out as the single largest monthly shipment in the dataset.

Query 5 — Total FOB by Continent (Exports)

Objective

Aggregate export FOB values at the continental level to understand the macro-geographic distribution of U.S. semiconductor outflows.

SQL Code

```
SELECT  continents,
        SUM(fobvalue) AS total_fob_by_c
FROM    exports_data
GROUP BY continents
ORDER BY total_fob_by_c DESC;
```

Result

Continent	Total FOB (USD)
Asia	17,270,967,059

North America	9,071,866,604
Europe	1,546,996,934
South America	989,522,020
Oceania	41,202,220
Africa	10,346,498

Key Design Choices

- The continents column was added during the data-cleaning phase as an enrichment field; this query validates that all six continents populated correctly.
- Asia dominates at ~\$17.3 B, nearly twice North America's ~\$9.1 B, reinforcing the strategic importance of the Asia-Pacific semiconductor supply chain for U.S. exporters.
- Africa's minimal share (~\$10.3 M) and Oceania's small share (~\$41 M) confirm those markets are negligible in the current dataset.

Troubleshooting Notes

Error 1054 (unknown column 'continent') was raised when the field was referred to without the trailing 's'. After correcting the column name to continents, the query returned 6 rows successfully.

Query 6 — Total FOB by Month (Exports)

Objective

Compute aggregate monthly export totals across all partners to reveal seasonal trends and identify peak shipment periods within the dataset.

SQL Code

```
SELECT    refMonth,
          SUM(fobvalue) AS monthly_exports
FROM      exports_data
GROUP BY  refMonth
ORDER BY  refMonth;
```

Result

Month	Total FOB (USD)
-------	-----------------

January	3,278,118,845
February	2,762,374,799
March	2,872,100,204
April	3,127,032,172
May	2,816,068,015
June	2,743,521,325
July	2,901,769,944
August	2,804,614,708
September	2,643,286,501
October	2,982,019,822

Key Design Choices

- Ordering by refMonth (alphabetically as stored) provides a natural calendar sequence for plotting.
- January and April are the two highest-volume months, suggesting possible fiscal-year or procurement-cycle effects.
- The relatively tight range (~\$2.6 B to ~\$3.3 B) indicates stable monthly demand without extreme seasonal swings.

Query 7 — Export Volatility by Partner

Objective

Measure the month-to-month variability of exports for each partner by computing both the average monthly FOB value and its standard deviation. High-volatility partners pose greater supply-chain and revenue-forecasting risk.

SQL Code

```
SELECT partnerISO,
       AVG(fobvalue) AS avg_exports,
       STDDEV(fobvalue) AS volatility
FROM   exports_data
GROUP  BY partnerISO
ORDER  BY volatility DESC;
```

Sample Results (top partners by volatility)

Partner	Avg Exports (USD)	Volatility (Std Dev)
China	619,638,176.9000	120,122,618.03
Mexico	853,484,687.0000	113,183,034.23
Malaysia	466,534,560.7000	100,794,533.26
Other Asia, nes	242,402,497.7000	77,487,308.98
South Korea	113,030,688.9000	28,899,547.39
Ireland	43,711,116.4000	23,131,708.03
Viet Nam	116,477,999.7000	19,587,161.49

Key Design Choices

- MySQL's built-in STDDEV() function computes population standard deviation, appropriate for describing the full distribution present in the dataset rather than estimating a sample.
- China tops the volatility ranking despite being the second-largest destination by total volume, suggesting significant month-to-month swings in Chinese demand.
- Mexico has the highest average exports but lower relative volatility than China, indicating more consistent shipment cadence.
- 142 partner rows are returned, covering the full range of export destinations in the dataset.

Query 8 — Market Share of the Top 10 Partners (Exports)

Objective

Combine absolute export totals with relative market-share percentages in a single result set, providing a self-contained ranking table suitable for executive dashboards and competitive benchmarking.

SQL Code

```
SELECT partnerISO,
       SUM(fobvalue)
       SUM(fobvalue) * 100.0 / (SELECT SUM(fobvalue)
                                FROM exports_data)
                                AS total_exports,
                                AS percent_share
FROM   exports_data
GROUP BY partnerISO
ORDER BY total_exports DESC
LIMIT 10;
```


Result

Partner	Total Exports (USD)	Market Share (%)
Mexico	8,534,846,870	29.50
China	6,196,381,769	21.42
Malaysia	4,665,345,607	16.13
Other Asia, nes	2,424,024,977	8.38
Viet Nam	1,164,779,997	4.03
South Korea	1,130,306,889	3.91
Brazil	842,205,357	2.91
Ireland	437,111,164	1.51
Canada	395,226,100	1.37
Singapore	384,311,392	1.33

Key Design Choices

- Multiplying by 100.0 (rather than 100) ensures floating-point division and two-decimal-place precision in the percentage column.
- The scalar subquery (SELECT SUM(fobvalue) FROM exports_data) is evaluated once and reused for every row, which is efficient given the small dataset size.
- The top three partners — Mexico, China, and Malaysia — together account for approximately 67% of all U.S. semiconductor exports, highlighting a highly concentrated demand side despite the lower HHI.
- Combining total_exports and percent_share in one query eliminates the need for post-processing joins and makes the result immediately usable for visualization.

Summary of Findings

The eight queries collectively paint a detailed picture of U.S. semiconductor trade flows:

- Import concentration (HHI ~0.70) is far higher than export concentration (HHI ~0.17), suggesting the U.S. buys from a narrow set of suppliers but sells to a broader global market.
- Mexico, China, and Malaysia dominate export destinations, together capturing roughly 67% of total FOB value.
- Asia as a continent receives over \$17 B in exports — nearly double North America's share — underlining the strategic weight of the Asia-Pacific region.
- Monthly export totals are relatively stable (~\$2.6 B–\$3.3 B), with January and April as peak months.

- China exhibits the highest month-to-month export volatility despite being the second-largest partner by volume, presenting forecasting challenges.
- The schema's continents enrichment column and clean partnerISO coding enabled efficient aggregation without additional joins, validating the data-engineering decisions made during the cleaning phase.